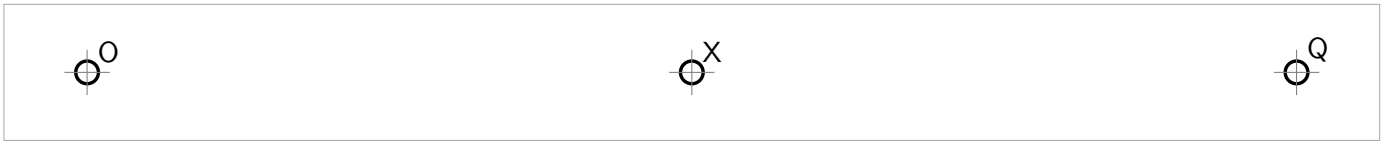
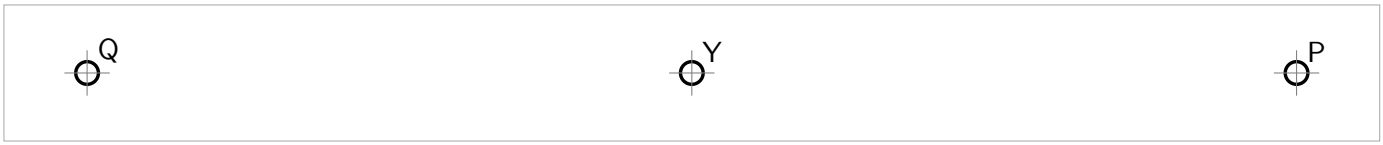
Adult fabrication template. Print at 100%, with no scaling. Rigid strips are 182 by 18 mm or 102 by 18 mm. Crosses mark hole centers. Hole diameter follows the chosen pivot or coaxial sleeve, not the printed circle. Prototype hardware and physical clearances are untested.



80 mm between neighboring hole centers



80 mm between neighboring hole centers

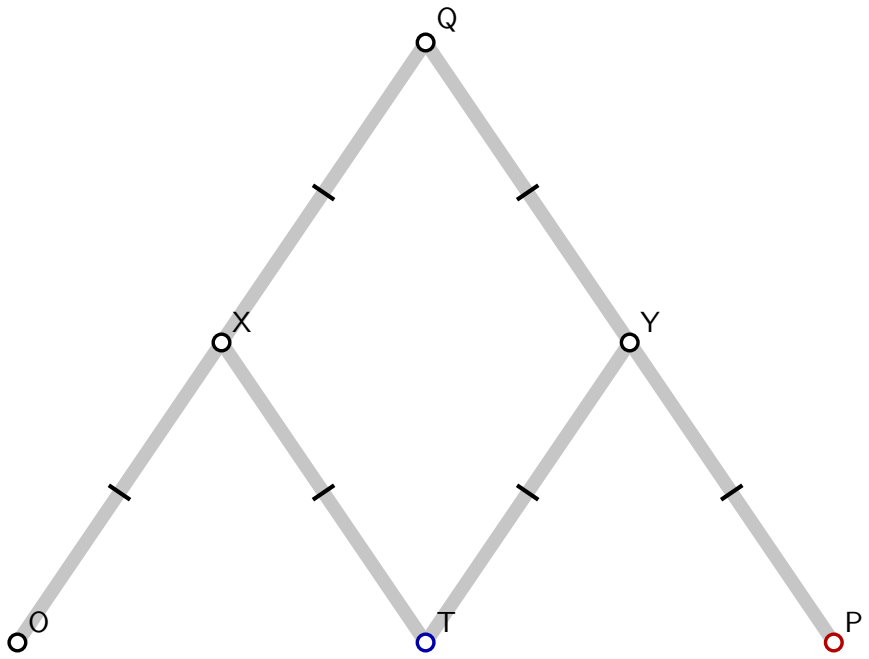


80 mm between neighboring hole centers



80 mm between neighboring hole centers

Adult assembly and print setup. Join O–X–Q as one long bar and Q–Y–P as the other. Join short bars X–T and T–Y. Fix O. X, Q and Y rotate. The T hinge must share its axis with the tracer pencil; P must carry its pencil on the P axis. Use spacers so all bars rotate freely.

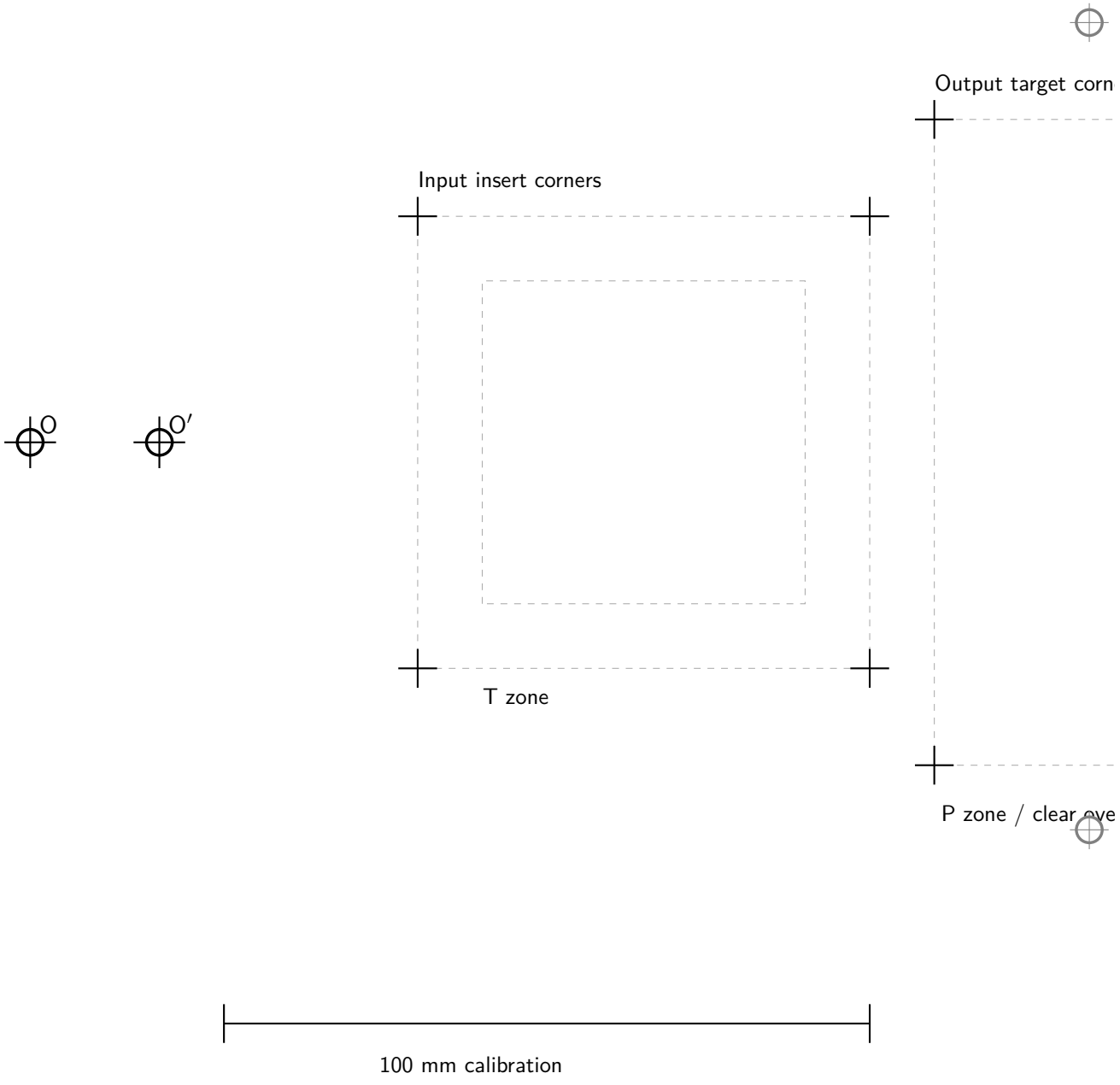


Keep X–Q–Y–T in the simple, noncrossed shape shown. Adult assembly must guard against collapse, sharp hardware, slipping anchors and off-axis tips. The line drawing is topology, not a tested stack of spacers or a sleeve specification.

Center the two-sheet mat on a board at least 450 by 400 mm. Match the paper edges and tape the backs. Cover the unprinted center seam with the bridge strip on page 5, aligning all four crosses. The joined sheets are 431.8 by 279.4 mm (nominally 432 by 279). Check the 100 mm line before use.

Before children use it, rehearse the whole T zone, both tracing directions and the anchor change. Check support, hand clearance, branch stability and coaxial tips. None of these physical tests has been done for this draft.

Left mat sheet. Butt the inside paper edge to the other sheet; attach the bridge strip over the seam.



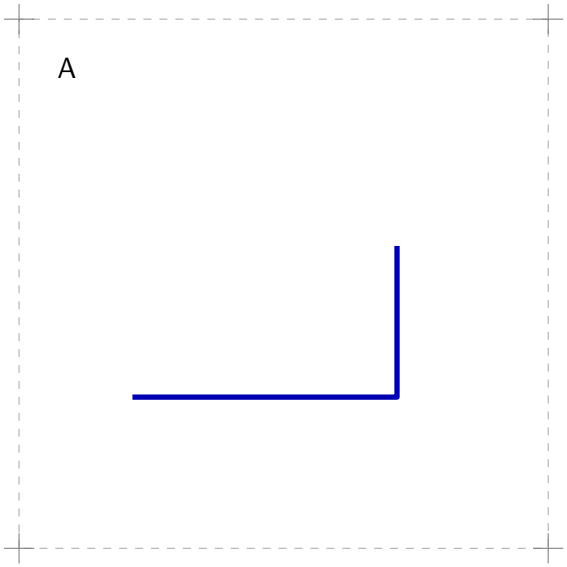
Print at 100%. Keep the sheet edges until after joining.

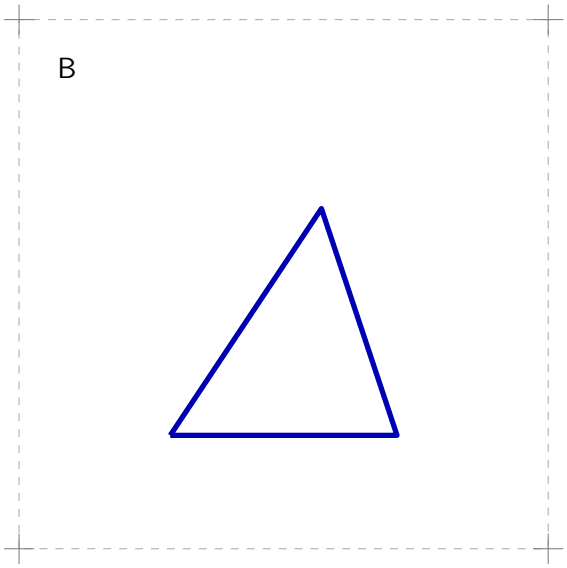
Right mat sheet. Butt the inside paper edge to the other sheet; attach the bridge strip over the seam.



Print at 100%. Keep the sheet edges until after joining.

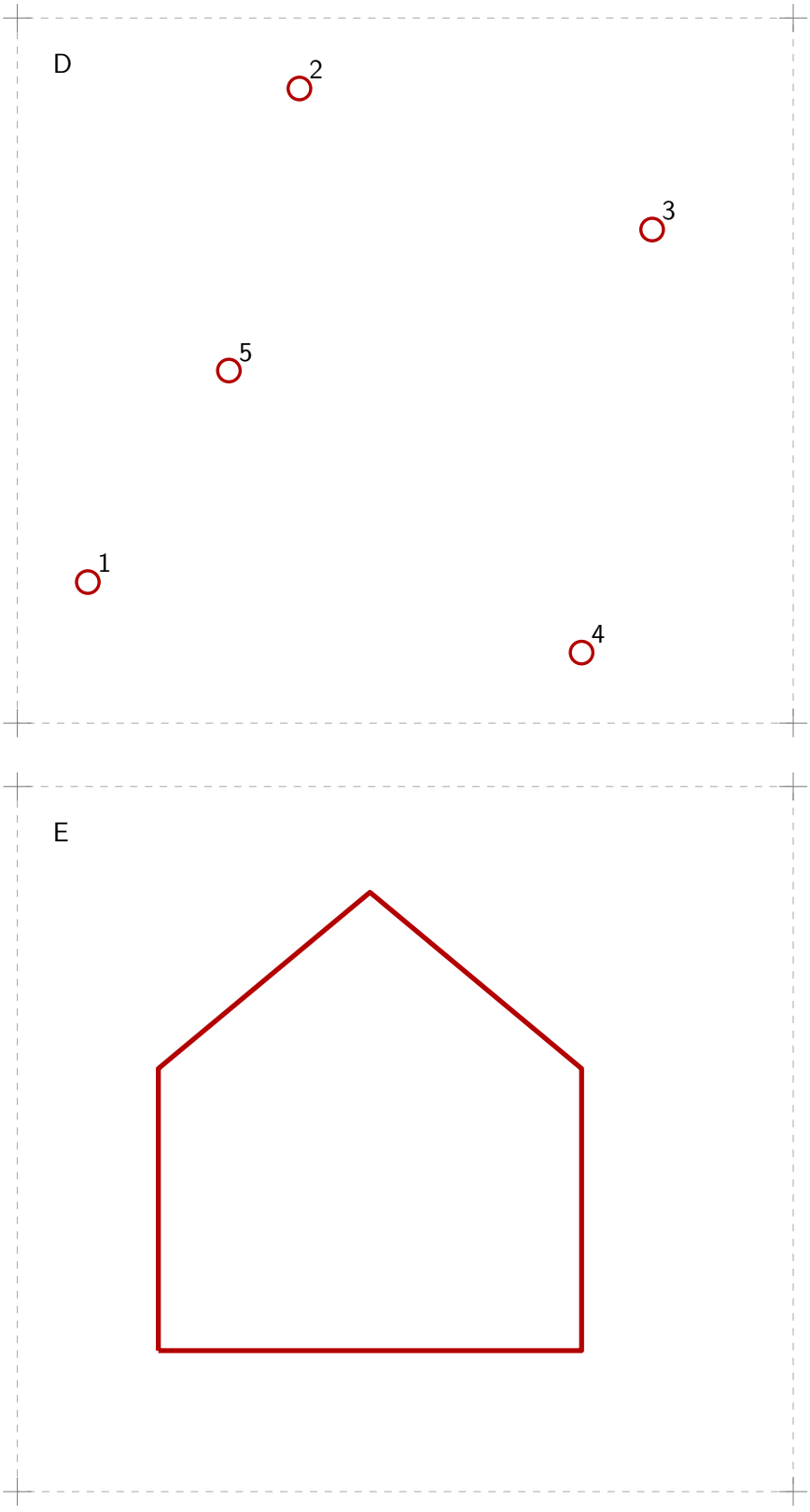
Cut the bridge on its outside rectangle. Align its four crosses with the joined sheets. Its center covers the unprinted seam. Cut inserts A and B on their 70 mm borders; place their corners on the mat's input corners.



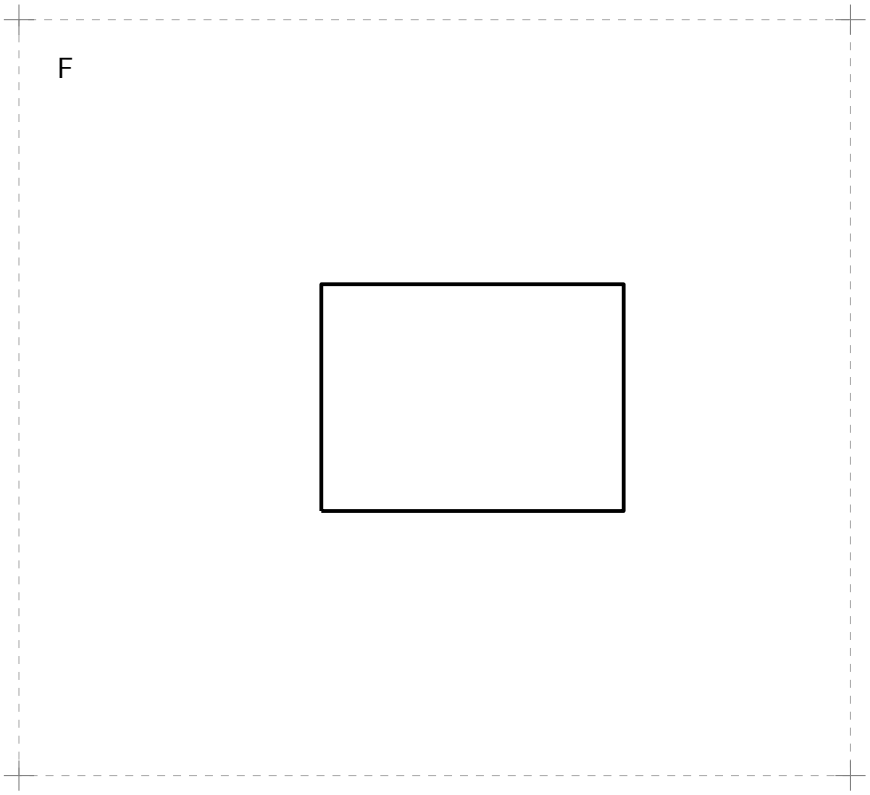


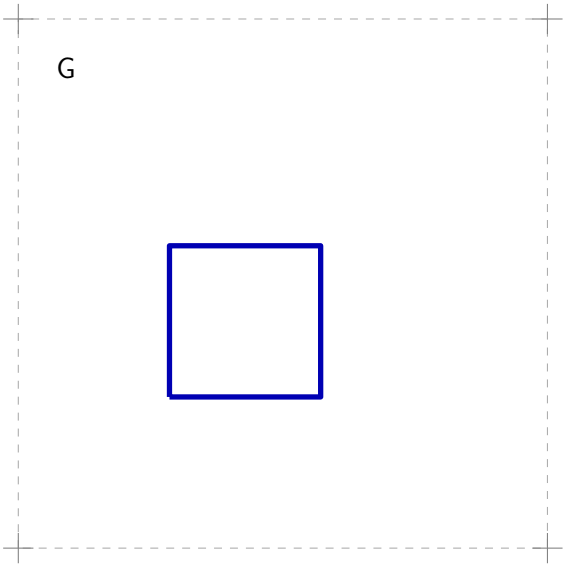


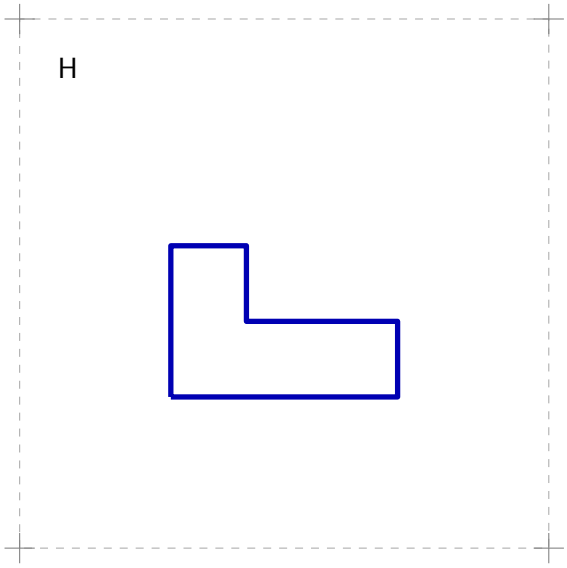
Cut the 110 by 100 mm targets. Place their corners on the output target corners of the joined mat, under a clear drawing overlay.



F goes on the output corners. G and H go on the input corners. Use one insert at a time and preserve each copy on its own overlay.

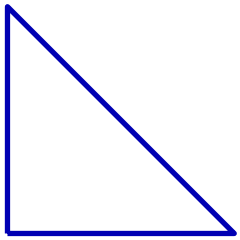






C, I, L and M go on the input corners. Keep M in its original place during the first anchor change. Move it only when the problem asks for a new place.

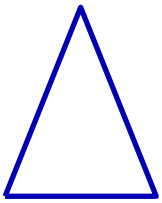
I



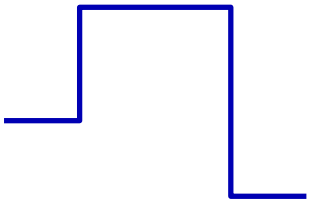
L



M



C

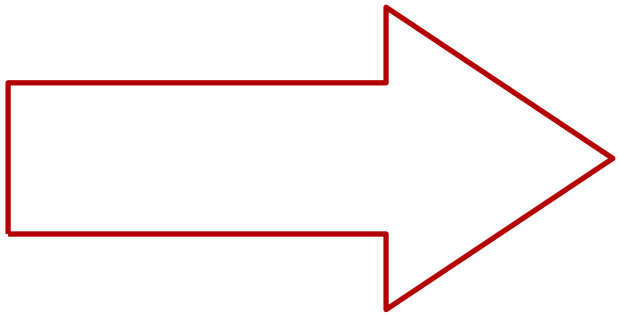


Place J or K on the output corners. An adult swaps the tip roles: move P while T draws.

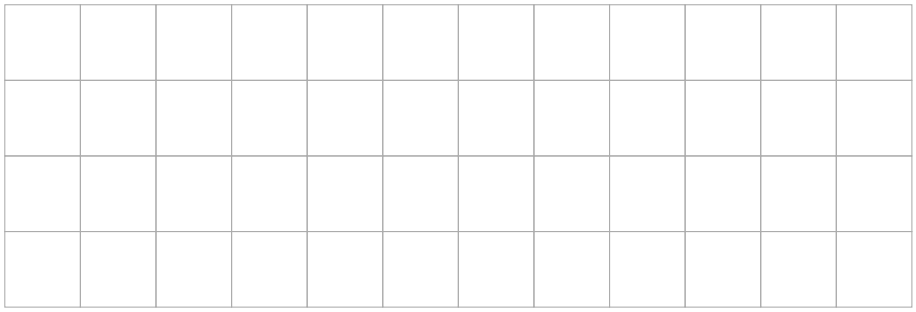
J

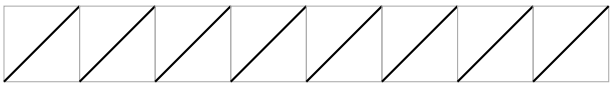


K



Cut 48 squares and 16 half-square triangles. Each square has 10 mm sides. Keep the triangles as separate pieces. These units are for one original and its copy at a time.





Also supply clear overlays large enough to cover both drawing zones, a 150 mm ruler, a straightedge, and removable tape. Pencil marks stay on overlays, so the mat and targets can be reused.

Material check per pair: one assembled machine; one board; one joined mat with bridge; inserts A–M; 48 squares; 16 triangles; ruler; straightedge; overlays. Physical fit and operation remain untested.